

46. What is a lambda function in Python?

Answer:

A lambda function is a small anonymous function (a function without a name). It is used when we need a simple function for a short time.

Syntax:

lambda arguments: expression

Example:

```
square = lambda x: x * x
```

```
print(square(5))
```

Output:

25

47. What is the difference between a normal function and a lambda function?

Answer:

Normal Function	Lambda Function
Has a name	Usually has no name
Uses def keyword	Uses lambda keyword
Can contain multiple statements	Contains only one expression
Used for larger tasks	Used for simple operations

Example:

Normal function:

```
def add(a, b):  
    return a + b
```

Lambda function:

```
add = lambda a, b: a + b
```

**48. What are *args and kwargs in Python?

Answer:

*args and **kwargs allow a function to accept a variable number of arguments.

***1. args**

Used to pass multiple positional arguments.

Example:

```
def add(*numbers):  
    print(numbers)
```

```
add(10, 20, 30)
```

Output:

```
(10, 20, 30)
```

****2. kwargs**

Used to pass multiple keyword arguments.

Example:

```
def info(**data):  
    print(data)
```

```
info(name="John", age=20)
```

Output:

```
{'name': 'John', 'age': 20}
```

49. What is recursion in Python?**Answer:**

Recursion is a process where a function calls itself to solve a problem.

Example:

```
def count(n):  
    if n == 0:  
        return  
    print(n)
```

```
count(n-1)
```

```
count(5)
```

Output:

```
5
```

```
4
```

```
3
```

```
2
```

```
1
```

Common uses:

- Factorial calculation
- Tree traversal
- Searching algorithms

50. What is the difference between local and global variables?

Answer:

Local Variable:

- Created inside a function.
- Can be used only inside that function.

Global Variable:

- Created outside a function.
- Can be accessed throughout the program.

Example:

```
x = 10 # Global variable
```

```
def show():
```

```
    y = 5 # Local variable
```

```
    print(y)
```

```
show()
```

```
print(x)
```

Output:

5

10